THEJESHWAAR SATHISHKUMAR

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SUMMARY

Undergraduate ML researcher specializing in deep learning, signal representations, and rigorous model evaluation. Experience across physiological time-series, EEG representation learning, multimodal seizure detection, and synthetic financial event detection. Interested in research and engineering opportunities spanning machine learning, neural engineering, and broader AI applications.

EDUCATION

Vellore Institute of Technology (VIT), Chennai | B.Tech. in Computer Science and Engineering **Expected 2027**

RESEARCH AND INDUSTRY EXPERIENCE

Mechanism-Routed Multimodal Seizure Detection | Lead Researcher **May 2026 – Present**

Manuscript under review at npj Digital Medicine

- Investigated clinical literature on wearable seizure monitoring to identify complementary signals beyond EEG, motivating a system integrating brain activity, heart activity, muscle activity, and movement.
- Audited an initial literature-guided multimodal model, tracing its poor performance to incomplete temporal coverage and ineffective fusion.
- Used sequential ablations to uncover early fusion limitations, a detrimental jaw-muscle feature, and wearable-data constraints not addressed by the initial approach. These findings informed a brain-context-guided design that dynamically weights three physiological pathways for each prediction.
- Evaluated the final model on 125 patients using the official SeizeIT2 patient-level split, achieving an AUROC of **83%** and outperforming both released baselines. Analysed pathway weights to examine how physiological evidence varied across the cohort and individual seizures.

ADHD Classification from EEG: Modeling and Evaluation Audit | Lead Researcher **Present**

- Reviewed 78 papers to investigate whether reported ADHD classification performance reflected meaningful neural patterns or methodological biases.
- Found that the literature's emphasis on high classification accuracy shaped my initial graph-based approach; discovered that selecting checkpoints on the test fold inflated its **92.19% accuracy by approximately 15 percentage points**, while a simpler 25-feature model achieved higher AUC (89.6% vs. 88.4%).
- Identified potential confounds in ADHD-versus-control classification, including residual 50 Hz artifacts and recording duration alone predicting group membership (**65.6% AUC**).
- Used these findings to guide approximately 40 systematic tests across feature representations, fusion strategies, and model architectures, requiring a prespecified minimum AUC improvement and randomly shuffling features to test whether gains depended on meaningful signal structure.
- Rebuilt the classifier using EEG covariance matrices mapped to Riemannian tangent space, achieving **93.34% AUC ± 1.49 percentage points** under subject-independent cross-validation, outperforming current state of the art.

Synthetic Financial Data & Event Detection | ML Research Intern, kyfr.ai **JUNE 2026 – AUG 2026**

- Built a synthetic financial data simulator generating daily transactions from demographic profiles and scripted life events, enabling model development when real-world financial data was limited.
- Developed and compared statistical, rule-based, gradient-boosting, and GRU-based detectors to identify rare financial events, evaluating their performance on synthetic data to inform approaches for real-world deployment.

Quantum-Hilbert Hybrid (QHH) EEG Representations | Lead Research Intern, CADS, VIT Chennai | **May 2025 – Nov 2025**

Manuscript under review at Frontiers In AI

- Investigated whether EEG phase information discarded by conventional power-based features provides complementary neural representations; developed a PennyLane variational quantum circuit encoding Hilbert-transform phase and amplitude to explore representation-level advantages over classical feature representations.

DESIGN SYSTEM PATENTS

- **Signal Reliability-Driven Multimodal Prosthetic Control System** — Patent No. 202641077871; issued June 2026.
- **Wearable EEG-Based Cognitive Monitoring System** — Patent Application No. 202641098121.
- **Hardware-Based Self-Healing IoT System** — Patent No. 202541081235; issued September 2025.
- **Railway Track Monitoring and Emergency Control System** — Patent Application No. 202641096039.

TECHNICAL SKILLS

Machine Learning: PyTorch, scikit-learn, CNNs, GRUs, Graph Neural Networks, Gradient Boosting, Multimodal Fusion, Domain-Adversarial Training

Signal Processing: MNE-Python, EEG Analysis, Hilbert Transform, Signal Representations, Connectivity Features

Research & Evaluation: Subject-Independent Cross-Validation, Ablation Studies, Data Leakage and Confound Analysis, Permutation Tests, Shuffled-Feature Controls

Tools: Python, NumPy, pandas, SciPy, PennyLane, FastAPI, Docker, Linux, Git